

IPBES nexus assessment

Chapter 5.4 – HEALTH, data management report 1

Multi-criteria decision analysis methodology for selecting response options

Code: IPBES_NXS_5.4.1

Version: v2.0.0

Project leader(s)

Hans Keune (hans.keune@uantwerpen.be)

Project contributor(s)

Purnamita Dasgupta (purnamita.dasgupta@gmail.com)

Rosemary A McFarlane (ro.mcfarlane@canberra.edu.au)

Carlos Das Neves (carlos.dasneves@efsa.europa.eu)

Pablo De La Cruz (pedelacruz@gmail.com)

Amanda Koltz (amanda.koltz@gmail.com)

Thijs Kuiken (t.kuiken@erasmusmc.nl)

Serge Morand (serge.morand@umontpellier.fr)

Rine Reuben (reubenrine@yahoo.com)

Johan Springael (johan.springael@uantwerpen.be)

Technical support unit

Tiff van Huysen (tiff.vanhuysen@un.org)

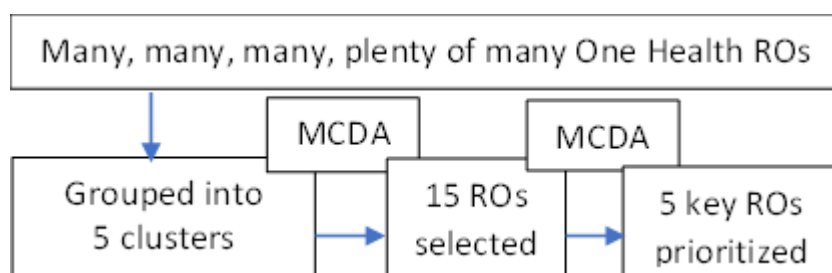
Data curator(s)

Hans Keune (hans.keune@uantwerpen.be)

Description

Multi-criteria decision analysis (MCDA) was used to identify, screen, and score response options (ROs) to select 15 ROs to describe briefly in the chapter and further select a subset of five ROs to describe in depth. Multi-criteria decision analysis is a non-monetary valuation method for simultaneously embracing, combining and structuring often incommensurable diversity: diversity of information (e.g., qualitative and quantitative data, as well as uncertainty), diversity of opinion (also among experts), diversity in actor perspectives (stakes), and diversity in assessment/decision-making criteria (Keune et al., 2013; Van Schoubroeck et al., 2019). Multi-criteria decision analysis also supports deliberation and helps to pave the way to decision-making and communication about decisions taken. The group ranking method was applied; this method was also used in IPBES Regional Assessment Report on Biodiversity and Ecosystem Services for Europe and Central Asia regarding health issues.

Process overview



Protocol

An initial set of 70 response options that address health issues and two or more nexus elements were selected by expert opinion (chapter authors) and through searching major groups of reports such as WHO Environment Health reports, IPBES assessments including the Workshop Report on Biodiversity and Pandemics, the report from the nexus assessment first Indigenous and local knowledge (ILK) dialogue, National Biodiversity Action Plans, The World Organisation for Animal Health (WOAH) Terrestrial and Aquatic Animal Health Codes, Health-Climate National Adaptation Plans (H-NAPS) and IPCC reports, including the Working Group II Assessment Reports. These response options exhibited a degree of regional or thematic duplication applicable at different scales. They were condensed to reduce duplication and to emphasise common themes and assessed against nexus criteria (see the nexus assessment chapter 5 introduction) using expert opinion informed by literature. [Note to reviewers: The initial set of 50 response options will be provided with the final draft of the report.]

Stepwise multi-criteria decision analysis process for CH 5.4 SOD

Step 1: Scoping potential response options

Based on reading, consultations, expertise of authors and a snowball process, chapter lead authors entered response options into an Excel table by 30 April 2023 to develop a long list of 50-100 potential response options.

Step 2: Ranking clustered response options

All response options were grouped into the following clusters:

1. Reduce Environmental Health Risks
2. ILPC and Traditional Health
3. Nature Based Solutions for Health
4. Healthcare and Health Services
5. Integrated Health responses

Each response option was judged on the basis of common criteria:

Food
Health
Climate
Water
Biodiversity
Socio-political
Economic Cost Benefits & Risks

Technical feasibility
Participation
Equity
Impact on global goals
Trade off
Transformational
Feasibility
Equity
Transformative change potential

Step 3: Group ranking

Within each cluster, each group member ranked their preferred top three response options per criterion (the other response options need not be scored for that criterion). Equal rankings were allowed, so possibilities are: 1-2-3, 1-1-1, 1-2-2, 1-1-2. The rankings of the different group members were put together. Based on the MCDA analysis, different weighting combinations were presented as input for group deliberation. Response options in which all group members ranked criteria four to eight within or outside the top three were not discussed further. The remaining response options were discussed individually: group members with divergent scores explain their reasons to the others. Group members had the opportunity to change their scores based on above discussion. Again MCDA was performed. From the five clusters, the top three response options from each cluster were selected to make up the top 15.

Step 4:

The final 15 response options, grouped in the same clusters, were put to the expert group for selecting the top five response options. Again, within each cluster, each group member ranked their preferred top three response options per criterion. This resulted in five response options based on MCDA, which was then discussed in the group to reach a final decision.

Reflection on the MCDA approach

The approach helped the chapter team account for all elements in a well-structured fashion and supported deliberation. The MCDA also facilitated a more focused and common understanding of the complexity of the response options and selecting those to be described in the chapter.

References

- Keune, H., Springael, J., & Keyser, W. D. (2013). Negotiated Complexity: Framing Multi-Criteria Decision Support in Environmental Health Practice. *American Journal of Operations Research*, 3(1), Article 1. <https://doi.org/10.4236/ajor.2013.31A015>
- Summary for policymakers of the regional assessment report on biodiversity and ecosystem services for Europe and Central Asia of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services: Appendix 2.8.* (n.d.). <https://doi.org/10.5281/zenodo.3237612>
- Van Schoubroeck, S., Springael, J., Van Dael, M., Malina, R., & Van Passel, S. (2019). Sustainability indicators for biobased chemicals: A Delphi study using Multi-Criteria Decision Analysis. *Resources, Conservation and Recycling*, 144, 198–208. <https://doi.org/10.1016/j.resconrec.2018.12.024>